

Biconnected Graph Triangulation

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A biconnected plane graph with three faces

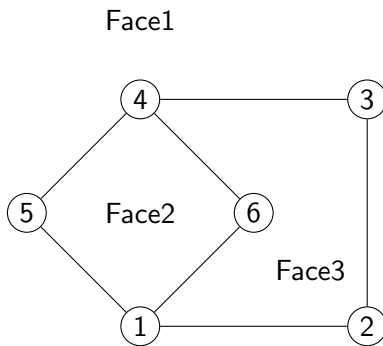
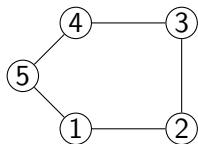
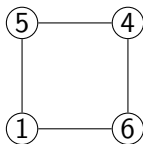


Figure: Root triangulation with safe root vertex 1. Chords from vertex 1 to all non-neighboring vertices (3, 4, 5) form a fan structure, ensuring no multi-edge conflicts with previously added chords.

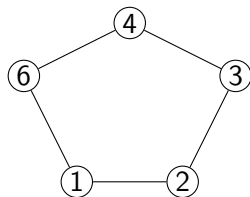
Faces of the Graph



(a) Face 1

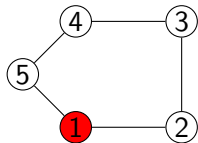


(b) Face 2

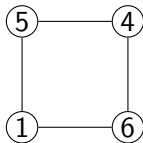


(c) Face 3

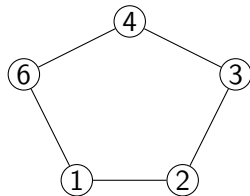
Find safe root of face 1



(a) **Face 1**



(b) Face 2

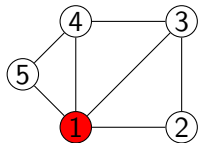


(c) Face 3

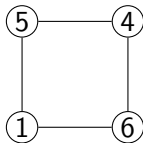
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6)$

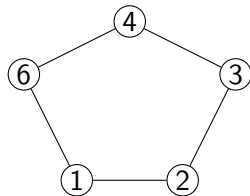
Root Triangulation of face 1



(a) **Face 1**



(b) Face 2

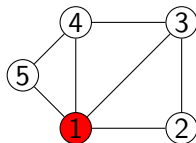


(c) Face 3

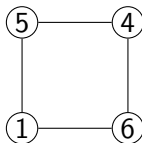
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4)$

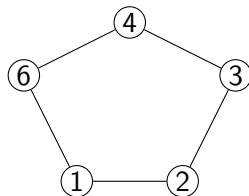
Now we move to face 2



(a) Face 1



(b) **Face 2**

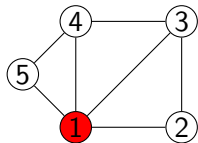


(c) Face 3

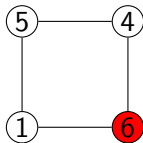
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4)$

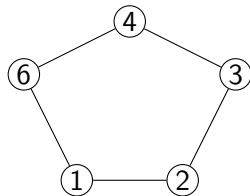
Find safe root of face 2



(a) Face 1



(b) **Face 2**

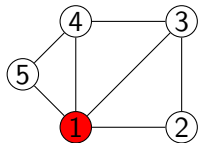


(c) Face 3

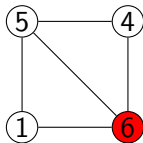
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4)**

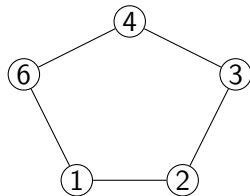
Root Triangulation of face 2



(a) Face 1



(b) **Face 2**

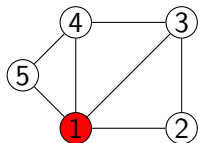


(c) Face 3

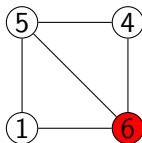
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

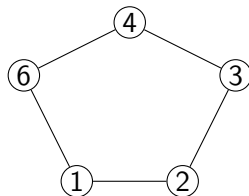
Now we move to face 3



(a) Face 1



(b) Face 2

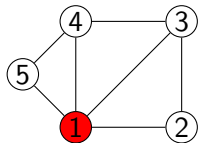


(c) **Face 3**

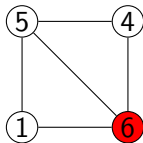
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

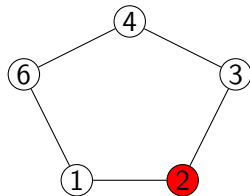
Find safe root of face 3



(a) Face 1



(b) Face 2

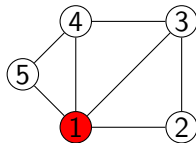


(c) **Face 3**

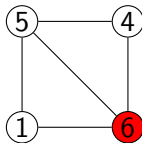
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(1,3), (1,4), (5,6)**

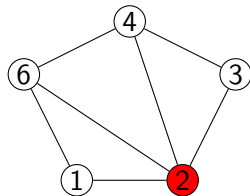
Root Triangulation of face 3



(a) Face 1



(b) Face 2

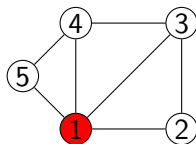


(c) **Face 3**

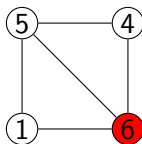
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6). (2,4), (2,6)

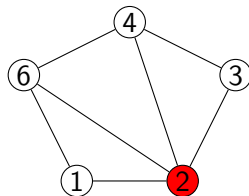
There is no other faces after face 3, so its a complete valid triangulation



(a) Face 1



(b) Face 2

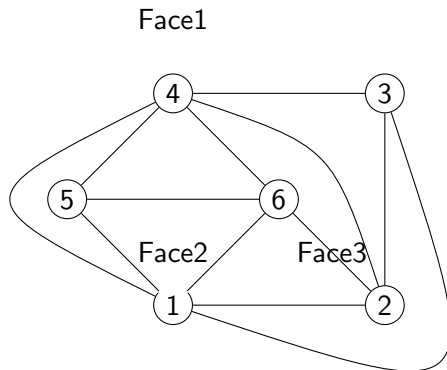


(c) Face 3

Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6). (2,4), (2,6)

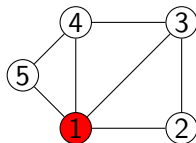
So the actual triangulation looks like this:



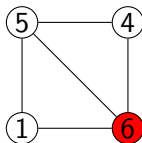
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6). (2,4), (2,6)

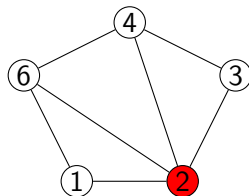
Now we will start flipping of Face 3



(a) Face 1



(b) Face 2

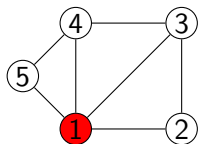


(c) Face 3

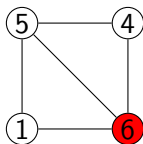
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6), (2,4), (2,6)$

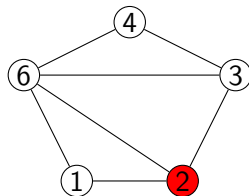
We can flip the (2,4) edge, as its opposite pair is not in the global edge set



(a) Face 1



(b) Face 2

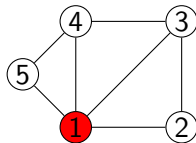


(c) Face 3

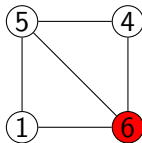
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6). (3,6), (2,6)

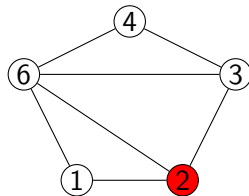
Its a new triangulation



(a) Face 1



(b) Face 2

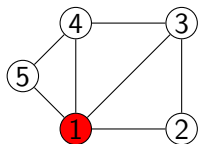


(c) Face 3

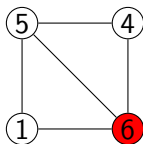
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6), (3,6), (2,6)

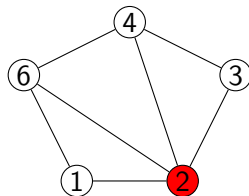
in the root also, cant flip (2,6) as (1,4) is present in the global edge set



(a) Face 1



(b) Face 2

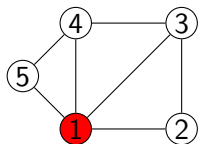


(c) Face 3

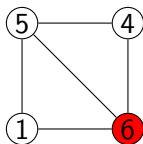
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6). (2,4), (2,6)

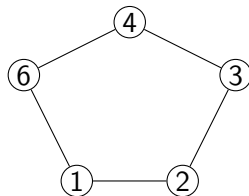
So, there is no more triangulations possible from here, so we backtrack to face 2



(a) Face 1



(b) Face 2

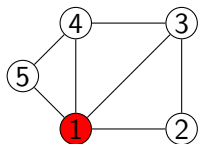


(c) Face 3

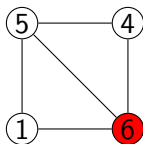
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6)$

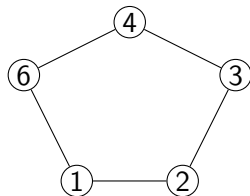
We can't flip the (5,6) edge as (1,4) is present in the global edge set



(a) Face 1



(b) Face 2

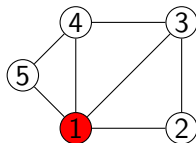


(c) Face 3

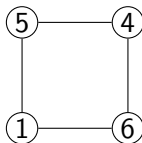
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4), (5,6)

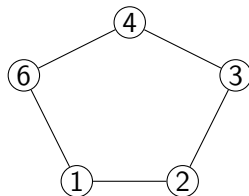
So we backtrack to face 1



(a) **Face 1**



(b) Face 2

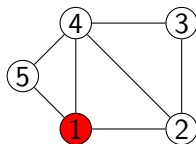


(c) Face 3

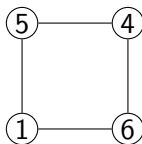
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (1,3), (1,4)$

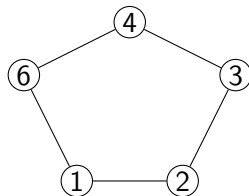
We can flip the (1,3) edge, as its opposite pair is not in the global edge set



(a) **Face 1**



(b) Face 2

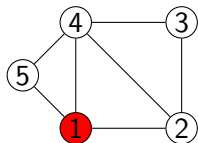


(c) Face 3

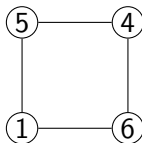
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (1,4)

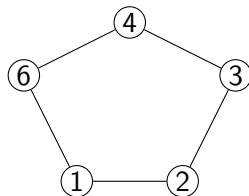
Now we move to face 2



(a) Face 1



(b) **Face 2**

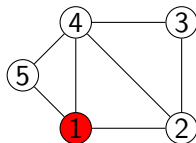


(c) Face 3

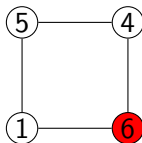
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (1,4)$

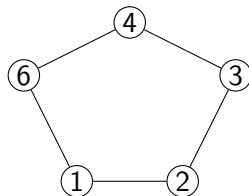
Find the safe root of face 2



(a) Face 1



(b) **Face 2**

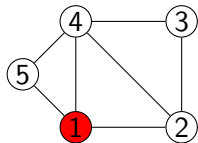


(c) Face 3

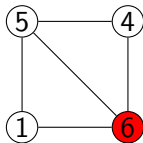
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (1,4)$

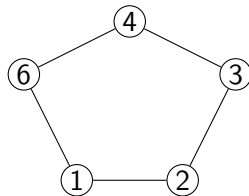
Root Triangulation of face 2



(a) Face 1



(b) Face 2

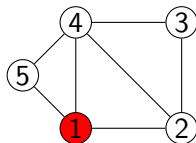


(c) Face 3

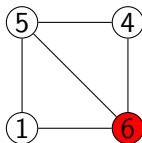
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (1,4), (5,6)

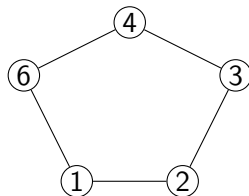
Now we move to face 3



(a) Face 1



(b) Face 2

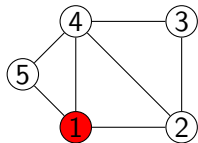


(c) **Face 3**

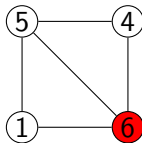
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (1,4), (5,6)$

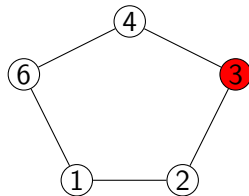
Find safe root of face 3



(a) Face 1



(b) Face 2

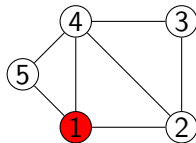


(c) **Face 3**

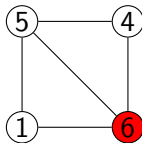
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (1,4), (5,6)

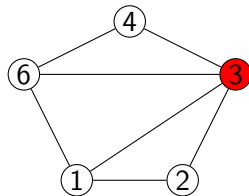
Root Triangulation of face 3



(a) Face 1



(b) Face 2

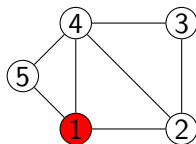


(c) Face 3

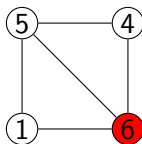
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2, 4), (1,4), (5,6), (1,3), (3,6)

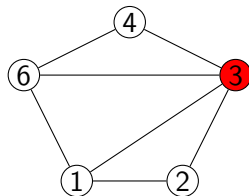
There is no face after face 3, so its a complete triangulation



(a) Face 1



(b) Face 2

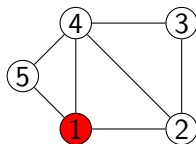


(c) Face 3

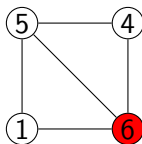
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2, 4), (1,4), (5,6), (1,3), (3,6)

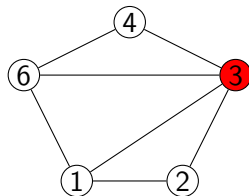
Cant flip (3,6), because opposite pair is (1,4), already exists



(a) Face 1



(b) Face 2

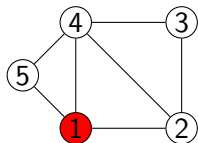


(c) Face 3

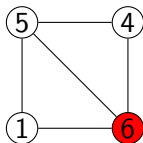
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2, 4), (1,4), (5,6), (1,3), (3,6)

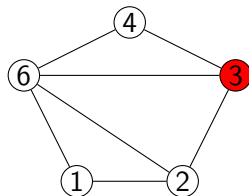
we can flip (1,3) to (2,6), its a new triangulation



(a) Face 1



(b) Face 2

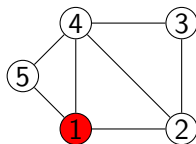


(c) Face 3

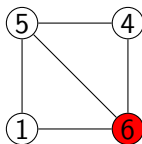
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2, 4), (1,4), (5,6), (2,6), (3,6)

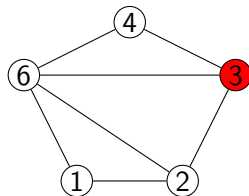
Cant flip (3,6), because opposite pair is (2,4), already exists



(a) Face 1



(b) Face 2

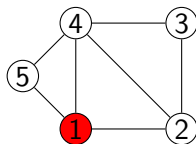


(c) Face 3

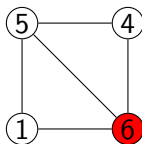
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2, 4), (1,4), (5,6), (2,6), (3,6)

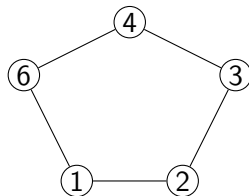
So, there is no more triangulations possible from here, so we backtrack to face 2



(a) Face 1



(b) Face 2

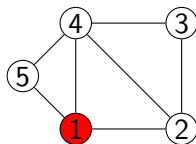


(c) Face 3

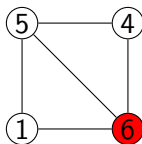
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (1,4), (5,6)$

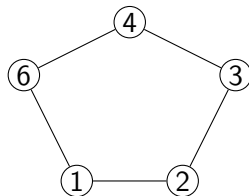
We can't flip the (5,6) edge as (1,4) is present in the global edge set



(a) Face 1



(b) Face 2

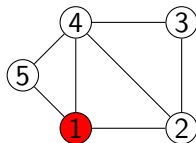


(c) Face 3

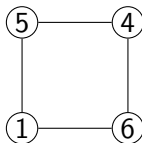
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2, 4), (1,4), (5,6)

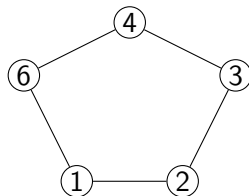
So we backtrack to face 1



(a) **Face 1**



(b) Face 2

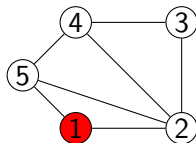


(c) Face 3

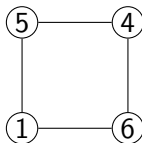
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (1,4)$

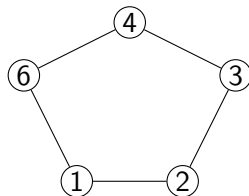
We can flip (1,4) to (2,5)



(a) Face 1



(b) Face 2

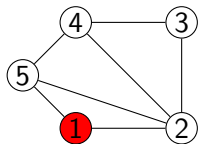


(c) Face 3

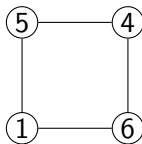
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), **(2,4), (2,5)**

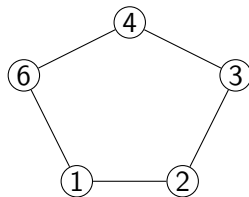
We can now move to Face 2



(a) Face 1



(b) **Face 2**

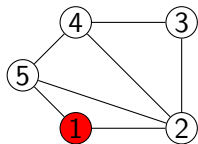


(c) Face 3

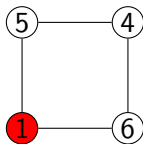
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5)$

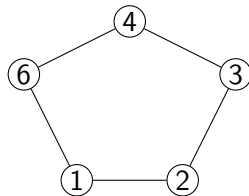
Find safe root of Face 2



(a) Face 1



(b) **Face 2**

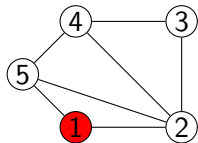


(c) Face 3

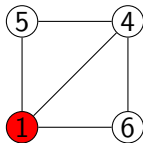
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5)$

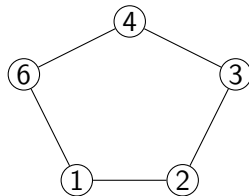
Root Triangulation of Face 2



(a) Face 1



(b) **Face 2**

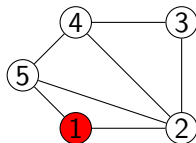


(c) Face 3

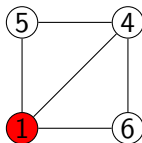
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5), (1,4)$

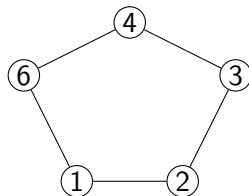
Now we move to Face 3



(a) Face 1



(b) Face 2

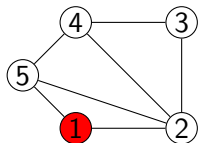


(c) **Face 3**

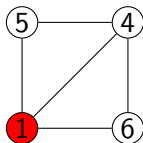
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5), (1,4)$

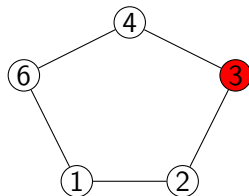
Find safe root of Face 3



(a) Face 1



(b) Face 2

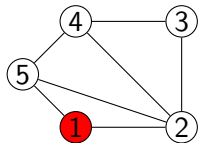


(c) **Face 3**

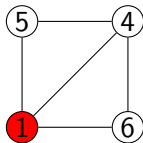
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5), (1,4)

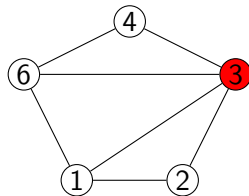
Root Triangulation of Face 3



(a) Face 1



(b) Face 2

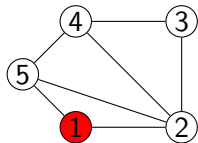


(c) **Face 3**

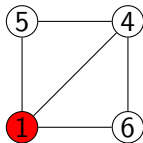
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5), (1,4), (3,6), (1,3)$

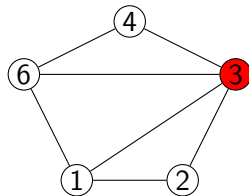
There is no more faces, so its a complete triangulation



(a) Face 1



(b) Face 2

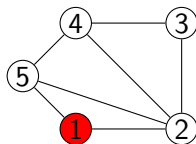


(c) Face 3

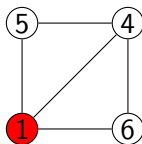
Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5), (1,4), (3,6), (1,3)$

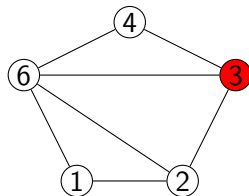
We can flip the (1,3) edge to (2,6) to get a different triangulation



(a) Face 1



(b) Face 2

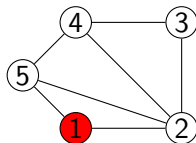


(c) Face 3

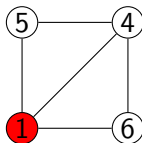
Global Edge Set:

(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5), (1,4), (3,6), (2,6)

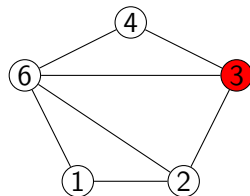
There is no more faces, so it's a complete triangulation



(a) Face 1



(b) Face 2



(c) Face 3

Global Edge Set:

$(1,2), (2,3), (3,4), (4,5), (1,5), (1,6), (4,6), (2,4), (2,5), (1,4), (3,6), (2,6)$

Continue....

We will be continuing this process until all faces are triangulated.